

Procurement refers to the process of obtaining goods and services from external sources to ensure that an organisation's operations can run smoothly and efficiently.

It includes identifying needs, selecting suppliers, negotiating contracts, purchasing items, and evaluating performance.

YOU MAY NEED TO ANSWER SOME OF THE FOLLOWING QUESTIONS FOR YOUR COMPANY RE THE CASE STUDY:

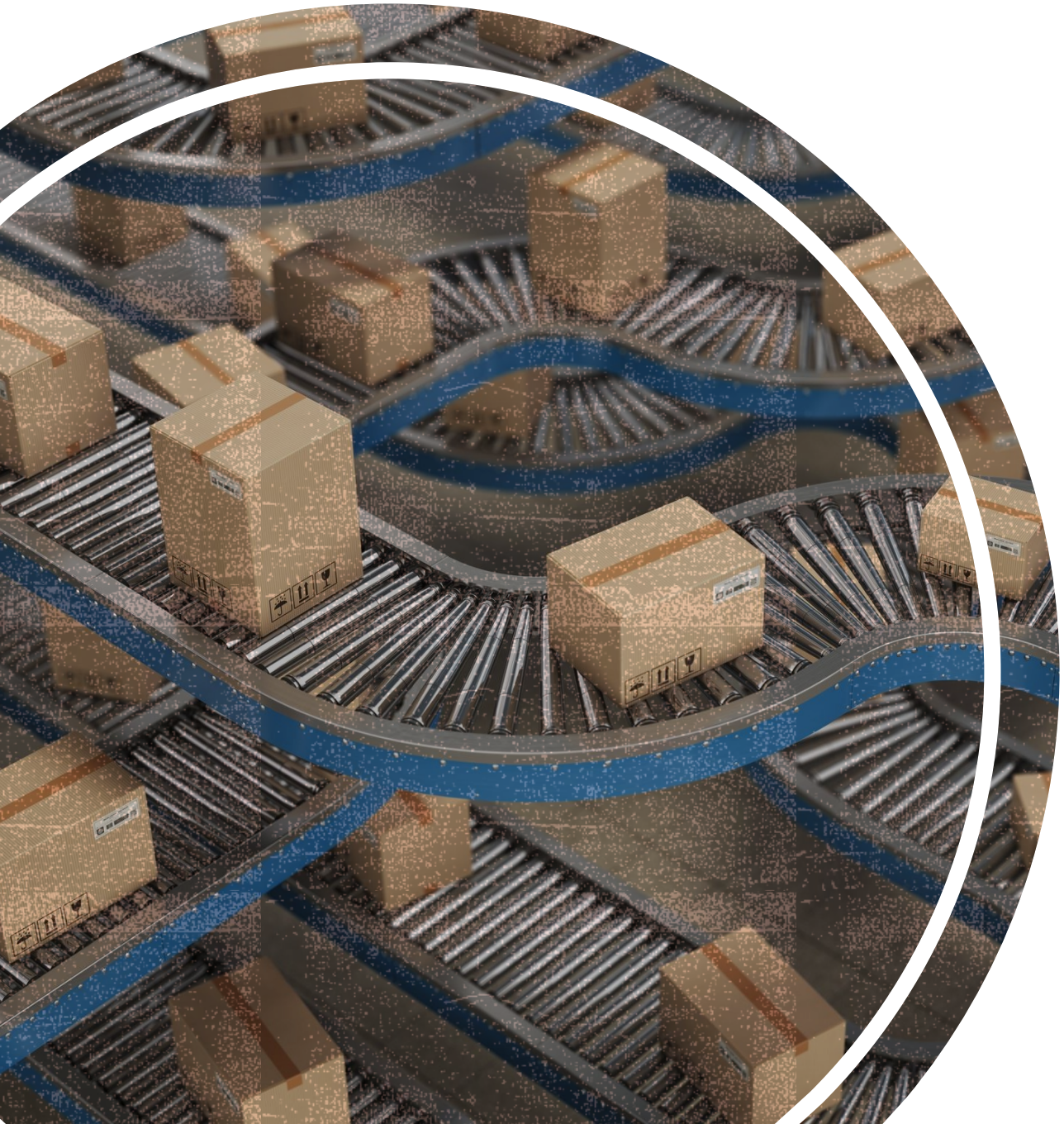
- How much of which material is needed and when?
- How much does the material requirement fluctuate?
- How accurately can fluctuations in demand be predicted?
- How much capital is tied up in material orders?
- How great is the risk that stocks will spoil or that customers will no longer want them?
- How well can materials be stored?
- What are the storage costs?
- How quickly can demand be met?
- How much effort is involved in determining demand?



THE ROLE OF PROCUREMENT IN BUSINESS

Procurement acts as a **strategic interface** between several functions:

- **Operations/Production:** ensures raw materials and components are available on time.
- **Finance**: manages budgets, cash flow, and cost efficiency.
- **Logistics:** coordinates inbound materials and warehouse management.
- **Marketing:** ensures materials for new product launches are ready.
- **Sustainability:** ensures suppliers meet environmental and social standards.



OBJECTIVES OF PROCUREMENT

- **Ensure availability of materials** to avoid production stoppages.
- **Minimize total costs** (purchase price, transportation, holding costs).
- **Ensure quality and reliability** of supplied goods.
- **Maintain flexibility** to adapt to market changes.
- **Support sustainability and compliance** with ethical sourcing standards.

DETERMINING PROCUREMENT REQUIREMENTS

Determining procurement requirements involves **calculating what materials are needed, how much, and when**. This is the foundation of material management and production planning.

There are 2 main methods:

(A) INVENTORY-DRIVEN METHODS

Inventory-driven methods determine procurement needs **based on stock levels**, without necessarily considering future demand directly.

Orders are triggered when stock drops to a specific level (reorder point).

Advantages:

- Simple and intuitive.
- Suitable for items with **stable consumption** and low value (C-items).
- Less data required.

Disadvantages:

- Not responsive to sudden demand changes.
- Risk of overstocking or stockouts if consumption varies.

Example:
A company orders printer paper when inventory reaches 2 boxes. The reorder point is purely based on historical consumption, not on future demand.

(B) DEMAND-DRIVEN METHODS

Procurement is based on **forecasted or planned demand**, typically linked to production schedules or customer orders.

Material requirements are derived from the production plan (Material Requirements Planning, MRP).

Advantages:

- Aligns closely with actual production needs.
- Prevents unnecessary inventory buildup.

Disadvantages:

- Requires accurate forecasting.
- Sensitive to demand fluctuations.

Example:
A car manufacturer orders specific engine components based on the number of cars planned for production next month.

SOME DEFINITIONS

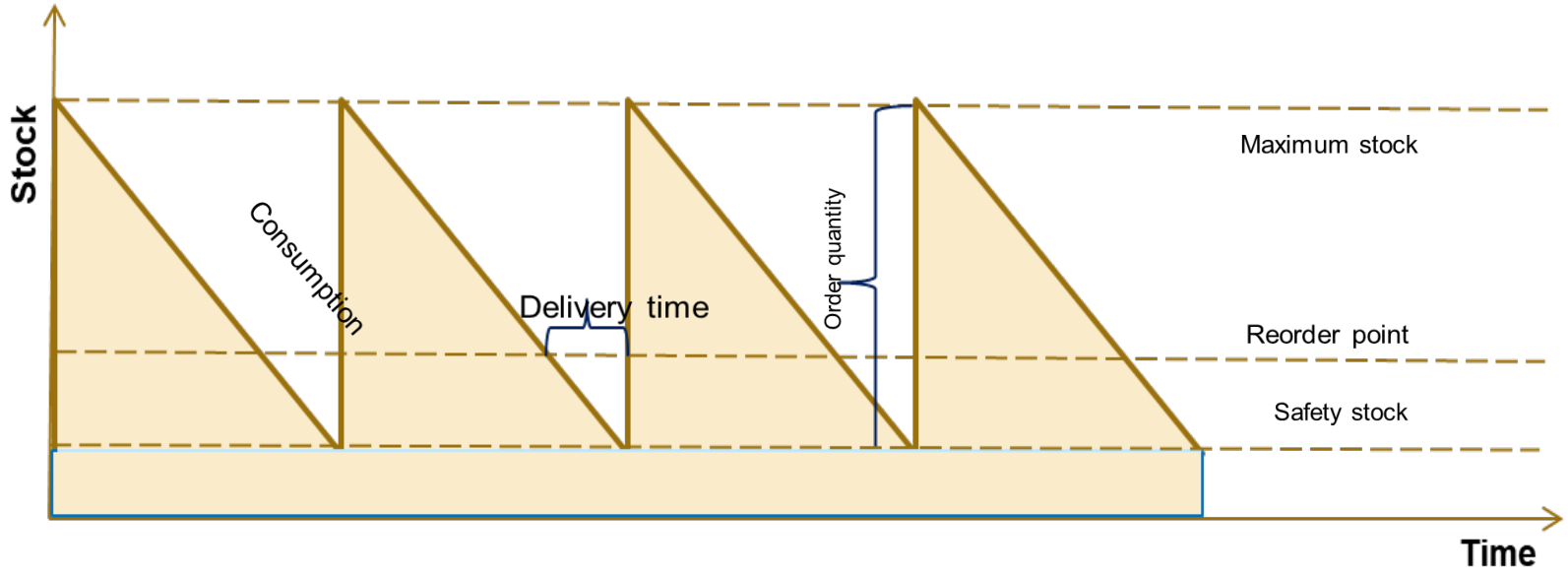
Stock: physical inventory in all warehouses

Available stock: stock freely available, i.e. warehouse stock minus reserved quantities plus any expected goods receipts

Safety (Buffer) stock: minimum quantity for emergencies

Reorder point: quantity at which an order is triggered

Maximum stock: maximum stock quantity



ORDER PLANNING USING INVENTORY-CONTROLLED METHODS

Procurement time: Time between determining a requirement and the point in time the goods are available. Procurement time includes all activities between recognising the need for materials and their availability in storage.

Components:

- **Order Preparation Time:** Internal approval, documentation.
- **Order Transmission Time:** Sending the order to supplier.
- **Supplier Lead Time:** Time for production and dispatch.
- **Transportation Time:** Shipping and customs.
- **Receiving & Inspection Time:** Quality check and storage.

Importance:
Accurate measurement of procurement time ensures that reorder points and safety stocks are properly set to prevent stockouts.

ORDER RHYTHM SYSTEM

Orders are placed at **regular intervals** (e.g., every two weeks), regardless of current stock level. The order quantity varies depending on consumption during the period. Mainly used in retail.

Useful if procurement dates:

- are specified by the supplier
- depend on circumstances for production/provision (e.g. season, routes)

Advantages

- Low planning and control effort, as no or hardly any stock checks are necessary.
- Bulk orders for similar materials are possible, resulting in better purchasing conditions.

Disadvantages

- Potential for costs due to missing quantities is relatively high à safety stock
- High storage costs
- High loss of value
- High capital costs

Example: A retailer orders replenishment every Monday based on the previous week's sales data.

REORDER POINT SYSTEM

Orders are placed **whenever stock reaches a predetermined level** (the reorder point). The quantity may be fixed or variable.

Useful when

- Orders are delivered reliably at all times
- Mainly in the manufacturing industry

Advantages

- More reliable assurance of material availability
- Lower safety stocks
- Cost-optimised order quantities possible

Disadvantages

- Stock checks required after each stock removal
- Individual orders, which may result in poorer purchasing/delivery conditions

Example: A car parts distributor orders 500 brake pads each time stock drops to 1,000 units.

Comparison Table System	Order Trigger	Order Quantity	Advantages	Disadvantages
Order Rhythm	Fixed time interval	Variable	Predictable schedule	Risk of shortages
Reorder Point	Stock level	Fixed or variable	Responsive to demand	Requires monitoring

DETERMINING THE OPTIMAL ORDER QUANTITY

$$EOQ = \sqrt{\frac{2 * D * S}{H}}$$

Where:
EOQ = Economic Order Quantity
D = Annual demand quantity (number of units)
S = Ordering cost per order
H = Holding cost per unit per year

The **Economic Order Quantity (EOQ)** model helps identify the order size that **minimizes total inventory cost**, balancing **ordering costs** and **holding costs**.

ASSUMPTIONS:



Constant procurement quantities



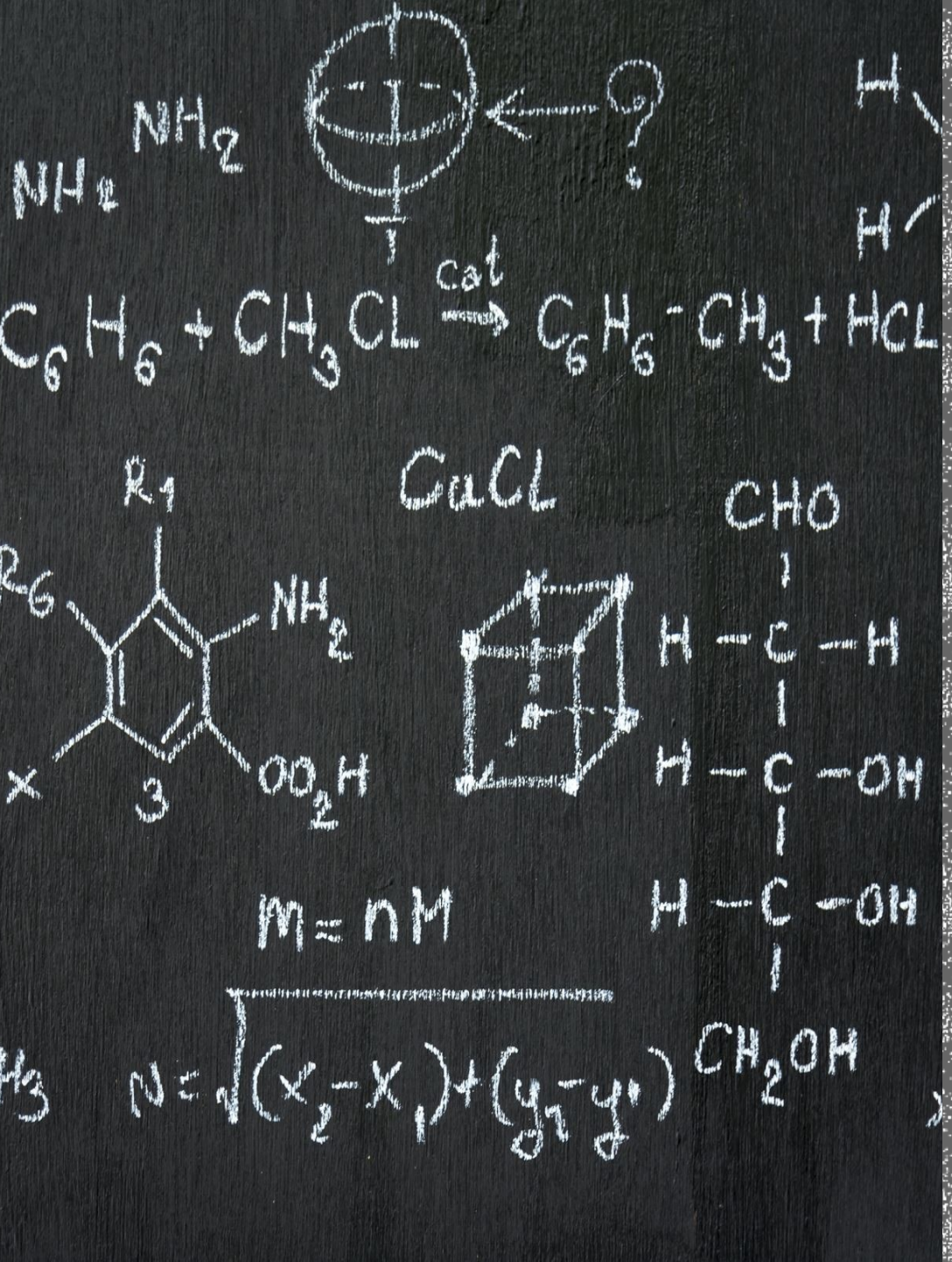
Stock removal rates remain the same



Cost prices are not dependent on the order quantity or the time of order



Fixed costs per order as well as interest and storage costs can be precisely determined and remain constant during the planning period



EXAMPLE:

Calculation

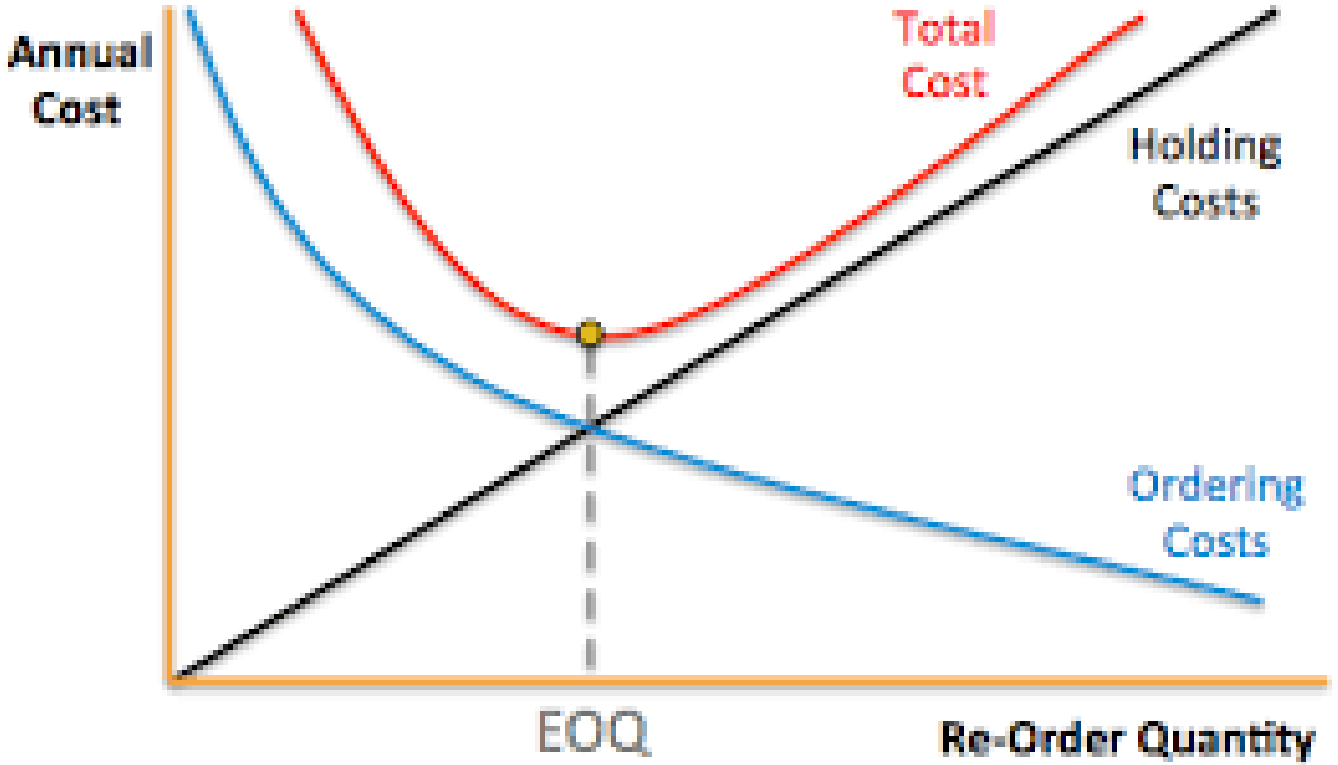
Annual demand = 10,000 units

Order cost = €50 per order

Holding cost = €2 per unit per year

EOQ = Square root of (2 x 10,000 x 50 / 2 = 707.1 units)

Ordering 707 units at a time minimizes the total cost of inventory.


Annual Cost
Total Cost
Holding Costs
Ordering Costs
EOQ
Re-Order Quantity

STOCK WASTAGE

Stock wastage refers to **losses in inventory due to damage, obsolescence, expiry, or poor handling**.

Types:

- **Physical wastage:** Spoilage, theft, or breakage.
- **Functional wastage:** Items become outdated or unnecessary.

Prevention Strategies:

- FIFO (First In, First Out)
- Periodic stock audits
- Digital tracking systems
- Supplier-managed inventory

IN-HOUSE PRODUCTION VS EXTERNAL PROCUREMENT

Aspect	In-House Production	External Procurement
Control	High – full oversight of quality & schedule	Low – depends on supplier
Cost Structure	High fixed costs, economies of scale	Lower fixed costs, flexible variable costs
Expertise	Internal knowledge required	Access to supplier know-how
Flexibility	Lower (requires capacity adjustments)	Higher (easier to scale)
Risk	Investment & capacity risk	Dependency & supply risk



STRATEGIC CONSIDERATION:

If a component is strategic to core competence, keep it in-house.

If it's **standardized or cost-sensitive**, consider outsourcing.

ABC ANALYSIS (PARETO PRINCIPLE)

ABC Analysis helps prioritize procurement management efforts by classifying inventory items according to their **annual consumption value** — a product of unit cost and annual usage.

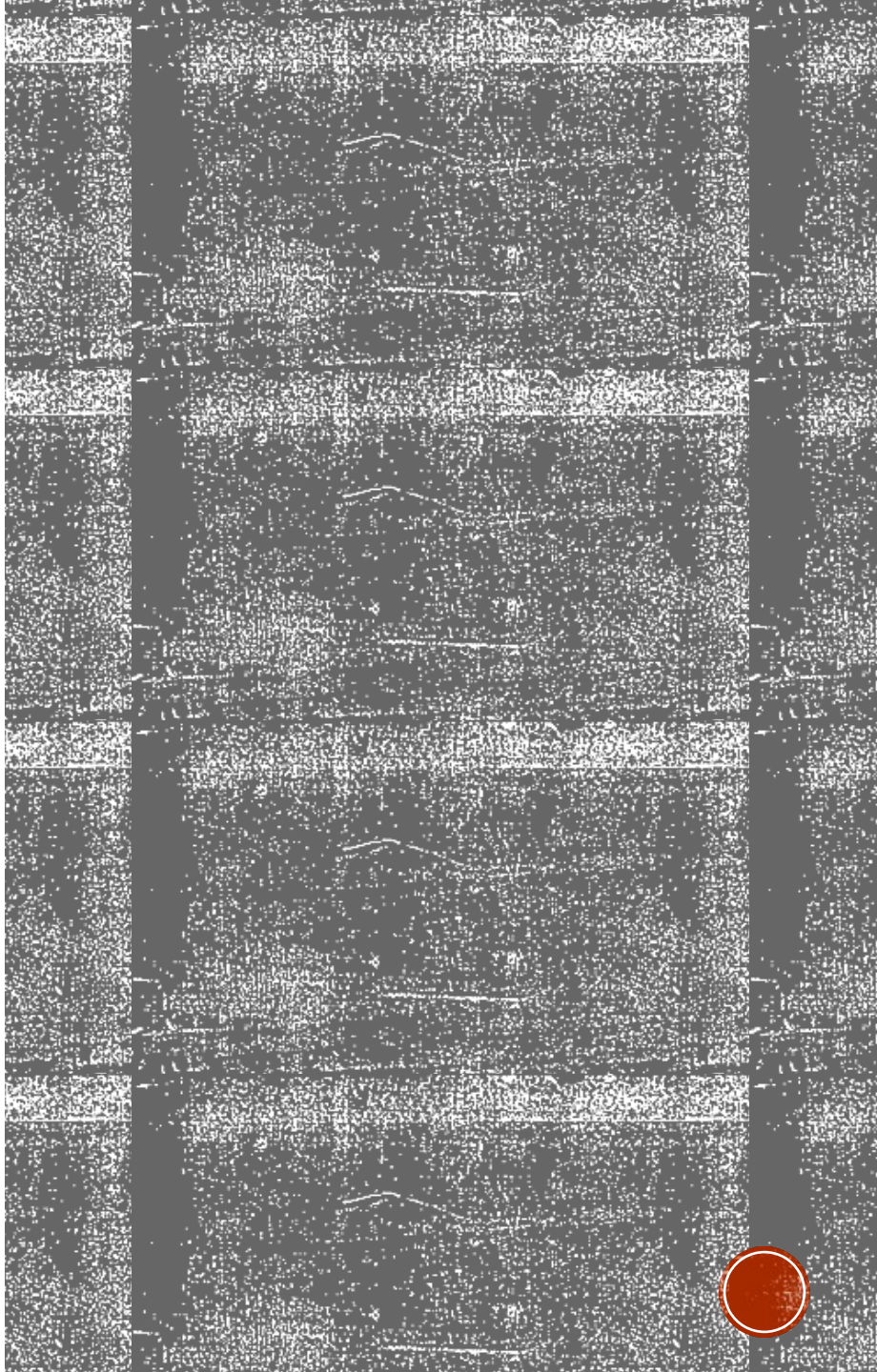
It applies the **Pareto Principle (80/20 rule)**:

- Roughly **20% of items** account for **80% of total inventory value**.
- The remaining items contribute less and can be managed with simpler control methods.

The Pareto effect implies that in procurement and inventory planning:

Priority should be given to A-goods, as this is where the greatest cost savings can be achieved
Methods: procurement market analyses, precise analysis of cost structures, determination of optimal order quantities and inventory levels, quality tracking, supplier controlling and evaluation

For C goods, on the other hand:
Only estimate the optimal order quantity roughly, keep higher safety stock levels, carry out less frequent stock checks, and only conduct limited procurement market analysis



PERFORMING AN ABC ANALYSIS:

1. Multiply the quantity by the value per unit à this gives the total value/total turnover per item

2. The individual items are sorted in descending order according to their total value (ranking)

3. The individual total values are added together à the sum corresponds to 100%

4. The percentage share of each product in the total value is determined

5. The total quantity of all goods is determined à the sum corresponds to 100%

6. The percentage share of each product in the total quantity is determined

7. The percentage shares of the total sum and the total quantity are accumulated in separate columns, i.e. the percentage values are added row by row and the subtotal is entered.

8. Categories are formed:

- **A goods:** approx. 70-80% of the total
- **B goods:** approx. 10-20% of the total
- **C goods:** only approx. 5-10% of the total

9. Graphical representation of the results

PARETO PRINCIPLE EXAMPLE – PERFORMING AN ABC ANALYSIS:

Description	Retail Price	Sales Quantity
Product 1	€ 890	4389
Product 2	€ 6,499	2984
Product 3	€ 189	7350

STEP 1 - MULTIPLY THE QUANTITY BY THE VALUE PER UNIT - THIS GIVES THE TOTAL VALUE/TOTAL TURNOVER PER ITEM

Product 1	€ 890	4389	€ 3,906,210
Product 2	€ 6,499	2984	€ 19,393,016
Product 3	€ 189	7350	€ 1,389,150
		14723	**€ 24,688,376**

STEPS 2-6

Rank		Price	No of Units	Total Value	Percentage of Units/Quantity	Percentage of Total Value/Revenue
1	Product 2	€ 6,499	2984	€ 19,393,016	20.27	**78.55**
2	Product 1	€ 890	4389	€ 3,906,210	29.81	**15.82**
3	Product 3	€ 189	7350	€ 1,389,150	49.92	**5.63**
			14723	€ 24,688,376	100.00	100.00

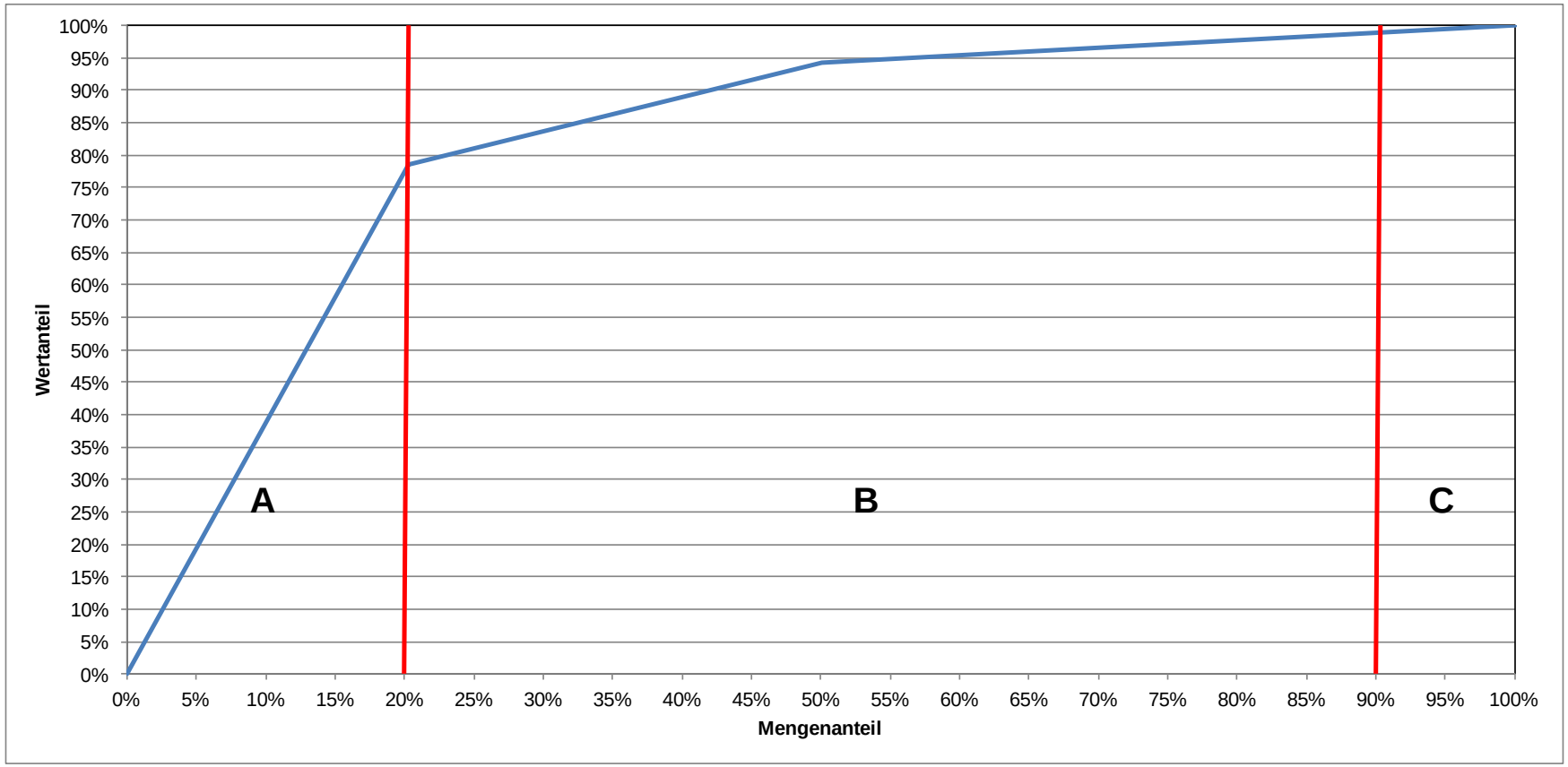


STEP 7

Description	Retail Price	Sales Quantity	Sales Value	% Share of Quantity	% Share of Revenue	Cumulative Share of Q	Cum Share of Rev	**Class**
Product 2	€ 6,499	2984	€ 19,393,016	20.27	78.55	20.27	78.55	**A**
Product 1	€ 890	4389	€ 3,906,210	29.81	15.82	50.08	94.37	**B**
Product 3	€ 189	7350	€ 1,389,150	49.92	5.63	100.00	100.00	**C**
		14723	**€ 24,688,376**	**100.00**	**100.00**			



GRAPH


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PARETO ANALYSIS QUESTION 1

PERFORM AN ABC ANALYSIS USING THE FOLLOWING VALUES

Category	Unit Price (€)	Volume (units)
Product 1	28.59	18,250
Product 2	497.36	10,385
Product 3	8.62	19,077
Product 4	19.78	35,725
Product 5	583.26	9,862
Product 6	161.46	9,735



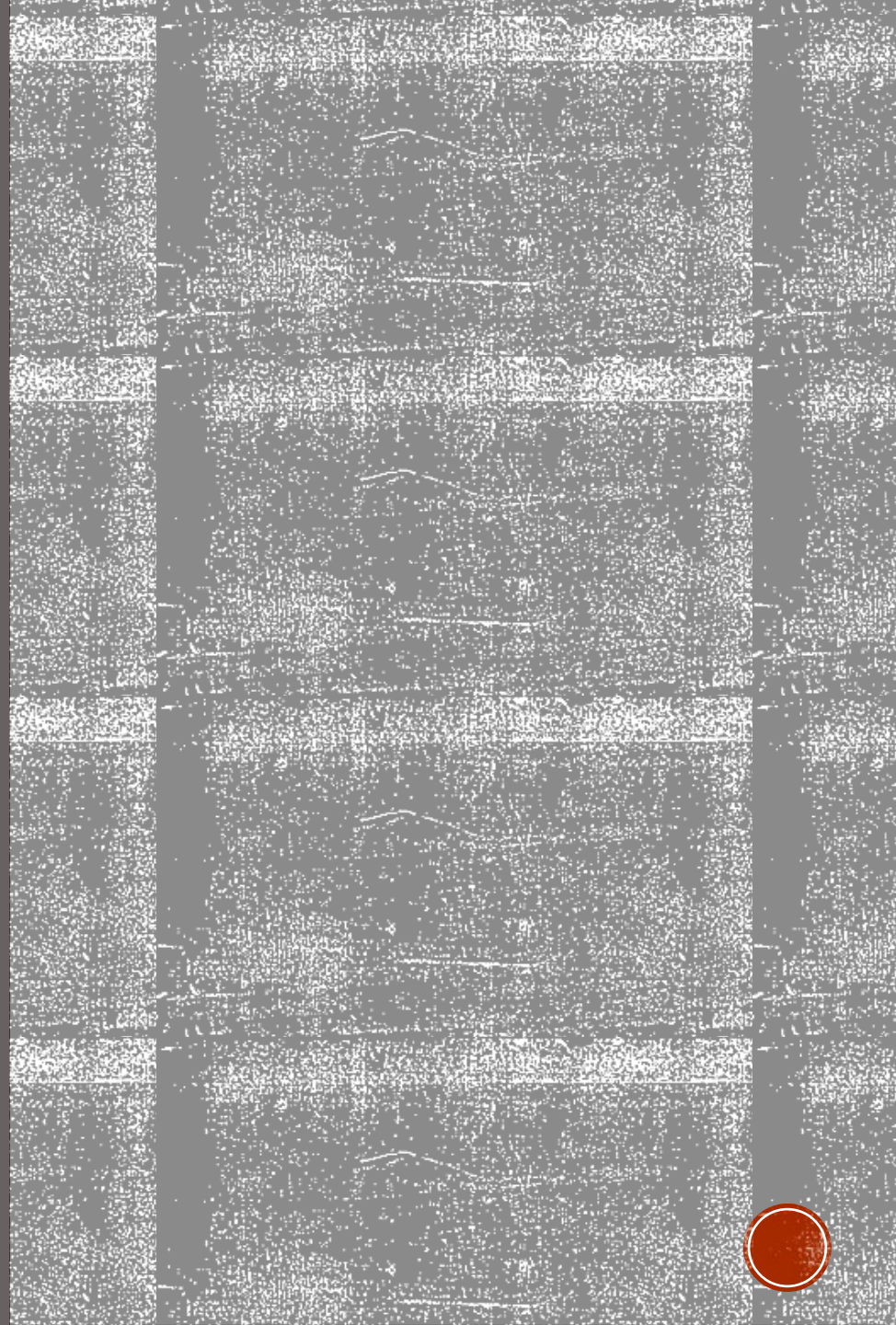
Solution – See Excel Spreadsheet on Ilias



Let’s consider a small manufacturing company that produces home appliances.

It manages **10 stock items**, each with a unit price and annual usage (units/year).

Total Annual Consumption Value = €222,000



Item	Unit Price (€)	Annual Usage (units)	Annual Consumption Value (€)	% of Total Value	Cumulative %	Class
A – Electric Motors	500	200	100,000	45.0%	45.0%	**A**
B – Steel Frames	120	400	48,000	21.6%	66.6%	**A**
C – Plastic Casings	40	600	24,000	10.8%	77.4%	**A**
D – Screws	1	15,000	15,000	6.8%	84.2%	**B**
E – Wires	5	2,000	10,000	4.5%	88.7%	**B**
F – Switches	10	800	8,000	3.6%	92.3%	**B**
G – Labels	0.5	10,000	5,000	2.3%	94.6%	**C**
H – Instruction Manuals	1	4,000	4,000	1.8%	96.4%	**C**
I – Packaging Boxes	0.8	5,000	4,000	1.8%	98.2%	**C**
J – Glue	2	2,000	4,000	1.8%	100%	**C**

HOW THE ANALYSIS IS CONDUCTED

Step 1: Calculate annual consumption value

Annual Value=Unit Price×Annual Usage

Step 2: Rank items by consumption value (descending).

Step 3: Calculate total consumption value and cumulative percentages.

Step 4: Classify items according to cumulative percentage:

- **A-items:** ~70–80% of total value (usually 10–20% of items)
- **B-items:** ~15–25% of total value
- **C-items:** ~5–10% of total value

INTERPRETATION

A-items (Electric Motors, Steel Frames, Casings):
High-value, low-quantity items. Require close monitoring, precise forecasting, and long-term supplier relationships.
→ Apply tight inventory control, **vendor contracts**, and **just-in-time** deliveries.

B-items (Screws, Wires, Switches):
Moderate impact on total value. Regular control and balanced stock levels.
→ Apply periodic reviews, **economic order quantity (EOQ)**, and standard safety stocks.

C-items (Labels, Manuals, Boxes, Glue):
Low-value, high-quantity items.
→ Use simplified control, **automatic reordering**, or **bulk purchasing** to reduce administrative costs.

MANAGERIAL ACTIONS TABLE

Class	Typical % of Items	Typical % of Value	Control Focus	Procurement Strategy
A	10–20%	70–80%	Very tight control	Negotiate contracts, close supplier cooperation
B	20–30%	15–25%	Regular control	EOQ, periodic review
C	50–70%	5–10%	Minimal control	Automate ordering, bulk purchase

STRATEGIC INSTRUMENTS: XYZ ANALYSIS

Concerned with the regularity of a process or fluctuations over time

- X materials: regular demand, standard materials

àProcurement based on demand

- Y materials: subject to regular fluctuations, e.g. due to seasonal fluctuations

àCertain stock levels are maintained, taking into account known periodicity

- Z materials: completely irregular

àProcurement only when required

Category	Demand Pattern	Forecast Accuracy	Action
X	Stable, predictable	High	Fixed contracts
Y	Fluctuating, seasonal	Medium	Periodic review
Z	Irregular, unpredictable	Low	Maintain safety stock

COMBINED ABC-XYZ MATRIX COMBINING BOTH ANALYSES SUPPORTS DIFFERENTIATED PROCUREMENT STRATEGIES.

	X	Y	Z
A	Secure contract with preferred supplier	Flexible contract	Safety stock for critical items
B	Regular review	Dual sourcing	Order on demand
C	Automate	Simplify	Order only when needed

SUPPLIER SELECTION & ASSESSMENT

1) Supplier Pre-Qualification: Initial screening based on capabilities, certifications, and references.

2) Supplier Evaluation: Assess based on weighted criteria:

- Price (30%)
- Quality (25%)
- Delivery reliability (20%)
- Sustainability (15%)
- Innovation & flexibility (10%)

3) Supplier Development: Continuous improvement, joint innovation projects.

4) Supplier Monitoring: KPI tracking, periodic reviews.

TRENDS IN PROCUREMENT

1. Digitalization and E-Procurement

- Use of ERP systems (SAP, Oracle) and online supplier platforms.
- Automation of purchase orders and invoicing.
- Predictive analytics for demand forecasting.

2. Sustainability and Ethical Sourcing

- Green procurement: minimizing environmental impact.
- Social compliance: ensuring fair labor practices in the supply chain.
- ESG (Environmental, Social, Governance) reporting.



3. Risk Management and Resilience

- Diversifying supplier base to avoid dependence.
- Nearshoring and reshoring post-COVID.
- Contingency planning and dual sourcing.

4. Globalization and Geopolitical Risks

- Managing international supply chain disruptions.
- Trade regulations, tariffs, and political risks.

5. Circular Economy

- Reuse, refurbish, recycle components.
- Closed-loop supply chains.

Electronic procurement - to reduce time-consuming paper-based procurement processes (average cost €100, with 80% of orders with a goods value of less than €1,000) through electronic processes (approx. €16) à cost reduction of 84%.

Specific advantages:

- 30% shorter delivery times
- 25% lower process costs
- 16% reduction in storage costs
- 8% reduction in purchase prices for direct goods
- 13% reduction in purchase prices for indirect goods

Particularly suitable for C-items due to low risk

More of a supporting role for B and A

